Sachin MB Gautham

Postdoctoral Researcher — Computational Soft Matter Physics & Machine Learning

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Summary

Interdisciplinary researcher with a Ph.D. in Chemical Engineering from IIT Madras and current postdoctoral experience at IISc Bengaluru. Expert in statistical thermodynamics and soft matter physics, with a specialized focus on multi-scale molecular modeling and machine learning. Proven expertise in utilizing exascale computing to solve complex challenges in polymer science and nanocomposite systems. I seek to establish an independent, interdisciplinary research laboratory that integrates molecular theory, high-fidelity simulations, and AI-driven optimization to accelerate the design of functional nanomaterials. Committed to academic excellence through high-impact research, securing external funding, and fostering a rigorous pedagogical environment.

Education

- **Ph.D., Chemical Engineering —** Indian Institute of Technology (IIT) Madras *Feb 2026*
- **M.E., Chemical Engineering —** Birla Institute of Technology and Science (BITS) Pilani *May 2019*
- **B.Tech., Chemical Engineering —** National Institute of Technology (NIT) Karnataka, Surathkal *Jul 2017*

Awards & Fellowships

- **ANRF National Post-Doctoral Fellowship (N-PDF)**, Anusandhan National Research Foundation (ANRF), Govt. of India — a competitive 2-year postdoctoral fellowship hosted by Prof. P. K. Maiti, Department of Physics, IISc Bengaluru (2026–2028).
- Won **First Prize for Best Poster Presentation** at PCCP-SPHERE 2025 (IISc) for the work titled *"Thickening of Liquids Using Copolymer Grafted Nanoparticles."*
- Awarded the **International Travel Scheme (ITS) Grant** by the DST, Govt. of India, to present doctoral research at the 12th International Liquid Matter Conference (2024) at the Max Planck Institute for Polymer Research, Mainz, Germany.
- Secured a high-performance computing allocation of **1,000,000 core-hours** (CNM 83298) at Argonne National Laboratory (Carbon Cluster), USA, for large-scale simulations of nanomaterial assemblies (2023).
- Recipient of a competitive **Institute Travel Grant** from IIT Madras to present Ph.D. research at a specialized materials symposium in the USA (2023).
- Awarded **First Prize for Best Paper Presentation** at ChemPlus 2023 for developing a deep learning framework titled *"Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning."*
- Completed Professional Industrial Training at JSW Steel (Hospet), gaining hands-on exposure to syn-gas production technologies and smelter furnace operations (2014).

Research Experience

Research Associate–I, Indian Institute of Science, Bengaluru, India *Jan 2026 – Present*

Advanced Atmospheric Water Harvesting (AWH) using Metal–Organic Frameworks (MOFs) and Covalent Organic Frameworks (COFs)

- Research conducted under the **ANRF National Post-Doctoral Fellowship (N-PDF)**, hosted by Prof. P. K. Maiti, Department of Physics, IISc Bengaluru (2026–2028).
- Served as a lead computational researcher on a high-impact, industrial-scale water harvesting project centered on CALF-20 and specialized COF architectures.
- Executed extensive Grand Canonical Monte Carlo (GCMC) simulations to rigorously evaluate multi-cycle water adsorption isotherms and hydrolytic stability of candidate MOF/COF structures.

- Developing Machine Learning Potentials (MLPs) to overcome the computational bottleneck of traditional force-field methods, accelerating water-uptake prediction speeds by orders of magnitude without sacrificing atomistic accuracy.
- Analyzed complex simulation datasets to derive structure–property relationships, directly informing the optimization of water capture efficiency and regeneration energy for industrial-scale deployment.
- Bridged the gap between fundamental molecular theory and engineering application through high-fidelity performance modeling and active participation in cross-functional technical discussions with experimental partners.

Postdoctoral Fellow, Accenture–IISc Bengaluru, India *Jul 2025 – Dec 2025*

IISc–Accenture Collaborative Initiative on Predictive Analytics for Polymeric Coatings

- Spearheaded a collaborative research effort between IISc and Accenture to develop a predictive machine learning framework for the next generation of industrial paint formulations.
- Developed high-fidelity molecular models for complex ternary systems involving poly(methyl methacrylate) (PMMA), polystyrene (PS), and titanium dioxide (TiO_2) nanoparticles to investigate interfacial physics and dispersion stability.
- Architected a data-driven model to map molecular-level descriptors (polymer architecture, grafting density, and nanoparticle loading) to macroscopic rheological and mechanical properties critical to the paint industry.
- Utilized the framework to predict key performance metrics such as viscosity profiles, pigment dispersion quality, and film-forming characteristics, significantly reducing the experimental trial-and-error cycle.
- Led the translation of fundamental soft matter physics into actionable insights for industrial scale-up; ongoing optimization of nanoparticle formulations yielded promising results in enhancing coating durability and opacity.
- Oversaw technical deliverables and regular progress briefings for industrial partners; the resulting computational frameworks received positive stakeholder validation for their alignment with commercial formulation benchmarks and real-world performance requirements.

Institute Research Fellow, Indian Institute of Technology Madras, Chennai, India *Aug 2020 – Jul 2025*

Deep Learning Frameworks for Potential of Mean Force (PMF) in Polymer-Grafted Nanoparticles (PGNs)

- Developed a novel deep learning architecture to map the PMF between PGNs, enabling rapid exploration of high-dimensional interaction landscapes that are traditionally computationally exhaustive.
- Engineered a robust data-generation pipeline that integrates advanced MD sampling techniques with neural networks to produce high-fidelity thermodynamic training datasets.
- Optimized deep learning models to predict interparticle interactions across a wide design space, including variations in grafting density, chain length, and solvent quality, ensuring the framework's versatility for different systems.
- Developed a phase diagram of a polymer nanocomposite based on molecular dynamics simulations.
- Disseminated findings through high-impact peer-reviewed publications and presentations at international forums, contributing a scalable computational paradigm to the field of soft matter physics.

Nanotetrapods Promote Polymer Flow Through Confinement-Induced Packing Frustration

- Investigated anisotropic self-assembly of nanoparticles in a polymer matrix using molecular dynamics simulations.
- Developed a computational framework to model various nanoparticle geometries in nanocomposites.
- Identified the correlation between nanoparticle shapes and the rheology of nanocomposites and elucidated the molecular mechanism behind shape-dependent viscosity.
- Created a coarse-grained model for non-spherical nanoparticles and predicted their self-assembly behavior using molecular dynamics simulations.
- Collaborated with Prof. Mithun Chowdhury's lab (IIT Bombay) to validate the coarse-grained model with experimental data.

Sequence-Defined Pareto Frontier of a Copolymer Structure

- Developed a computational framework for designing copolymer sequences using genetic algorithms and molecular dynamics simulations.
- Analyzed structure–property correlations, such as radius of gyration, from a large collection of copolymer sequences.

Researcher, BITS Pilani, Goa, India *Aug 2017 – May 2020*

Nano-Enhanced Sulfonated Poly(Arylene Ether Sulfone) Composite Membranes and Their Characterization

- Designed and synthesized sPBI–SPAES with TiO_2 nanocomposite membranes for enhanced desalination properties.
- Cast and characterized reverse osmosis membranes incorporating TiO_2 nanocomposites.
- Focused on improving desalination efficiency through the integration of nanocomposites in the membrane structure.
- Conducted in-house synthesis of reverse osmosis membranes to optimize material properties.
- Enhanced mechanical properties of nanocomposite membranes compared to pristine membranes.
- Investigated the performance of TiO_2 nanocomposite membranes in reverse osmosis processes.
- Contributed significantly to publishing peer-reviewed journal articles based on the research.

Independent Research Projects

1. **Understanding Switching Dynamics of Tantalum Oxide and Titanium Oxide based Memristive Devices**
   Grant No. CNM55595, Agency: Argonne National Laboratory (ANL) — Allocation: 500,000 CPU hours on the *Carbon* supercomputer (2022).
2. **Polymer Directed Self-Assembly of Polymer Grafted Nanoparticles**
   Grant No. CNM 83298, Agency: Argonne National Laboratory (ANL) — Allocation: 1,000,000 CPU hours on the *Carbon* supercomputer, Center for Nanoscale Materials (2024).

Mentorship Activities

1. Mentored Mr. Abhishek for M.S. thesis on *"Glass Transition in Ion-Containing Polymers"* (2021–2022).
2. Mentored Ms. Sayani Karmakar for M.S. thesis on *"Molecular Dynamics Simulation of Ion Transport in Polymers"* (2022–2023).

Teaching Experience

Teaching Assistant, Indian Institute of Technology Madras *Aug 2021 – Jul 2024*

- Topics included thermodynamics, unit operations laboratory, introduction to molecular simulations, statistical thermodynamics, and thermodynamics of fluid mixtures.
- Prepared course material including computer codes, lectures, homework, and practice problems.
- Led weekly laboratory and/or problem-solving and discussion sessions.
- Supervised undergraduate and graduate students in final projects and weekly homework.

Management & Services

- Member, American Institute of Chemical Engineers (AIChE), American Physical Society (APS), and Society of Polymer Science, India (SPSI).
- Member, Organizing Committee, ChemPlus school-cum-symposium, IIT Madras, Chennai, India (2022).

Journal Publications

1. Sarkar J, **Gautham S M B**, Ali F, Madhusudanan M, Yadav H, Datta A, Patra T K, Chandran S, and Chowdhury M. Nanotetrapods Promote Polymer Flow Through Confinement-Induced Packing Frustration. *Nature Communications* **16**, 8550 (2025).
2. Adhya P, **Gautham S M B**, Patra T K, Kaushal M, and Mondal T. Thickening of Liquids Using Copolymer Grafted Nanoparticles. *Soft Matter* **21**, 4156–4164 (2025).
3. **Gautham S M B**, and Patra T K. Deep Learning Potential of Mean Force Between Polymer Grafted Nanoparticles. *Soft Matter* **18**, 7909–7916 (2022).
4. Bale A, **Gautham S M B**, and Patra T K. Sequence-Defined Pareto Frontier of a Copolymer Structure. *Journal of Polymer Science* **60**, 2100 (2022).

5. Venkatachalam K R, **Gautham S M B**, and Krishnan J N. Nano-Enhanced Sulfonated Poly(Arylene Ether Sulfone) Composite Membranes and Their Characterization. *High Performance Polymers* **34**, 1193 (2022).
6. Venkatachalam K R, **Gautham S M B**, and Krishnan J N. Blend Membranes of Sulphonated Poly(arylene ether sulphone) and Sulphonated Polybenzimidazole and Their Characterization for Desalination Applications. *Bulletin of Materials Science* **47**, 1 (2024).
7. Venkatachalam K R, **Gautham S M B**, and Krishnan J N. Desalination Characteristics of New Blend Membranes Based on Sulfonated Polybenzimidazole and Sulfonated Poly(Arylene Ether Sulfone). *Polymer Bulletin* **80**, 7805–7824 (2023).

Manuscripts Under Preparation and Review

1. **Gautham S M B**, Banerjee R, and Maiti P K. Optimizing Water Harvesting in TTBA COF: An Advanced NVT+W Approach (2026).
2. **Gautham S M B**, and Maiti P K. Tuning the Viscoelasticity of Paint Formulations using Copolymer Grafted Nanocomposites (2026).
3. **Gautham S M B**, Banerjee R, and Maiti P K. Molecular Mechanisms of Water Adsorption in Functionalized Covalent Organic Frameworks (2027).
4. **Gautham S M B**, and Patra T K. Machine Learning Guided Design of Multi-Component Copolymer Nanocomposites for Industrial Coatings (2027).

Thesis

1. **Ph.D. Thesis**: Microstructure–Rheology Correlations in Polymer Nanocomposites, Indian Institute of Technology Madras (2026).
2. **M.Tech. Thesis**: The Synthesis of Polymer Nanocomposite Membranes for Desalination, Birla Institute of Technology and Science, Pilani (2019).
3. **B.Tech. Thesis**: Preparation and Characterization of Immobilized Bacteria in Sodium Alginate, National Institute of Technology Karnataka, Surathkal (2017).

Talks and Poster Presentations

1. **Gautham S M B**, and Patra T K. Thickening of Liquids Using Copolymer Grafted Nanoparticles. RSC Physical Chemistry Chemical Physics (PCCP) SPHERE, IISc Bengaluru, India (October 2025).
2. **Gautham S M B**, and Patra T K. Effects of Nanoparticle Shape on the Rheology of Polymer Nanocomposites. Liquid Matter Conference (LMC), Mainz, Germany (September 22–27, 2024).
3. **Gautham S M B**, and Patra T K. Non-linear Rheology of Polymer Grafted Nanoparticle Solutions. Complex Fluids (CompFlu), Chennai, India (December 2023).
4. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. Sustainable Polymer Society of India (SPSI) MACRO, Guwahati, India (December 2023).
5. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. CECAM MD60, Bengaluru, India (September 2023).
6. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. ChemPlus, IIT Madras, Chennai, India (March 2023).
7. **Gautham S M B**, and Patra T K. Predicting Polymer Grafted Nanoparticles Assembly using Deep Learning. American Physical Society (APS) March Meeting, Las Vegas, Nevada, USA (March 5–10, 2023).
8. **Gautham S M B**, and Patra T K. Predicting GNPs Assembly using Deep Learning. Complex Fluids (CompFlu), Kolkata, India (December 19–21, 2022).